CLEAN AGENT FLOW DIAGRAMS WITH CODE EXPLANATIONS

DIAGRAM 1: MAIN AGENT PIPELINE FLOW

```
                    USER INPUT
             state.last_user_message = text
             conversation_turns += 1
                        │
                        ▼
        CONTROL LAYER (apply_control)
        • Check explicit human request
        • Check attempts > 3
        • Calculate escalation score
        • Check correction loops
                        │
        ┌───────────────┼───────────────┬───────────┐
       YES                              NO         NULL
        │                               │           │
        ▼                               ▼           ▼
  CONTROL ACTION               Continue normal flow
        │
   ┌────┴────────┐
HANDOVER       RESET
   │             │
   ▼             ▼
 HUMAN         SOFT RESET
 HANDOVER      ask_rephrase()
 (Line         (Line 3575-3583)
 3560-3569)          │
                     ▼
        state.just_reset = True
        state.awaiting_new_issue=True
        Preserve all entities
        Clear effort counters
                     │
                     ▼
        Check state.intent == None OR
        state.awaiting_new_issue == True
        (Line 3589-3595)
```

```
└──────────────────────┬──────────────────────┘
                       │
                       ▼
┌─────────────────────────────────────────────┐
│ INITIAL CASE EXTRACTION                     │
│ • extract_full() mode                       │
│ • Detect: intent, entities,                 │
│   sentiment, language                       │
│ • update_state()                            │
│ • update_case(mode="global")                │
│ (Line 3597-3611)                            │
└──────────────────────┬──────────────────────┘
                       │
                       ▼
┌─────────────────────────────────────────────┐
│ Check: just_reset == True?                  │
│ (Line 3617-3625)                            │
└────────────┬─────────────────┬──────────────┘
             │                 │
            YES                NO
             │                 │
             ▼                 ▼
┌────────────────────┐         │
│ REVERIFY LOW       │         │
│ CONFIDENCE FIELDS  │         │
│ (< 0.80)           │         │
│ ask_reverify_      │         │
│ low_confidence()   │         │
└────────────┬───────┘         │
             │                 │
             └────────┬────────┘
                      │
                      ▼
┌─────────────────────────────────────────────┐
│ Check: missing_fields?                      │
│ (Line 3631-3636)                            │
└────────────┬─────────────────┬──────────────┘
             │                 │
            YES                NO
             │                 │
             ▼                 ▼
┌────────────────────┐         │
│ ASK MISSING        │         │
│ ask_missing()      │         │
│ (Line 3633-3636)   │         │
└────────────┬───────┘         │
             │                 │
             └────────┬────────┘
                      │
                      ▼
╔═════════════════════════════════════════════╗
║ build_verification_response()               ║
║ • Create summary                            ║
║ • Ask user: "Is this correct?"              ║
║ (Line 3642-3646)                            ║
╚═════════════════════════════════════════════╝
                      │
```

```
DECISION ENGINE
decide(state, text)
(Line 3803-3806)

      YES                 NO
CORRECTIONS          COMPLETION

(See local correction diagram below)
```

DIAGRAM 2: CONTROL LAYER - ESCALATION DECISION

```
apply_control(state, text)
(Line 3289-3365)

CHECK 1:          CHECK 2-6:        CHECK 7:
EXPLICIT          PATTERN           ESCALATION
HUMAN             DETECTION         SCORE
REQUEST           (Line             (Line 3341)
(Line             3320-3339)
3301-3305)

YES               TRIGGER           CALCULATE
                  sentiment=        FORMULA:
                  "high"            (A*0.4) +
                                    (C*0.3) +
                                    (S*0.3)

KEYWORDS:         SET sentiment
• "human"         to "high"
• "agent"
• "real person"

ACTION:
IMMEDIATE
HANDOVER
(no questions)

                  Score >= 0.75?

                  YES                    NO

                  HANDOVER          Score >= 0.55?
                  (reason:
                  "escalation_      YES              NO
                  score_
```

```
critical")
                         SOFT RESET            NO ACTION
                         (if not               CONTINUE
                         just_reset)           NORMAL FLOW
```

```
ADDITIONAL CHECK: CORRECTION LOOP (Line 3355-3363)

if (awaiting_correction AND attempts >= 2 AND NOT just_reset):
    ▼
SOFT RESET
(reason:
"correction_loop")
```

DIAGRAM 3: HUMAN HANDOVER FLOW

```
HUMAN HANDOVER TRIGGERED
(Line 3560-3569)
        │
        ▼
state.escalate(reason)
• Log escalation reason
• Mark case as escalated
(Line 3562-3564)
        │
        ▼
COLLECT DATA TO SEND TO HUMAN AGENT

• state.entities: All extracted fields
• state.entity_confidence: Reliability scores
• state.sentiment: User's emotional state
• state.language: Detected language
• state.conversation_turns: Number of exchanges
• state.attempts: Failed field extractions
• state.correction_count: User corrections made
• state.last_user_message: Latest input
        │
        ▼
SEND HANDOVER MESSAGE TO USER

"I understand you're having difficulty with this process.
 I'm connecting you to a human agent who has full access
```

```
  to your account and can provide personalized assistance.

  Please hold while I transfer you..."

action: "handover"
              │
              ▼
┌──────────────────────────────────────────────────────────┐
│ AGENT PIPELINE STOPS                                     │
│ • No more automated responses                            │
│ • Human agent takes over                                 │
│ • Case marked as: ESCALATED / IN_PROGRESS                │
└──────────────────────────────────────────────────────────┘
              │
              ▼
┌──────────────────────────────────────────────────────────┐
│ HUMAN AGENT RECEIVES:                                    │
│                                                          │
│ ✓ Full conversation history                              │
│ ✓ All extracted data (name, email, issue, etc.)          │
│ ✓ Confidence scores for each field                       │
│ ✓ Sentiment analysis (high frustration?)                 │
│ ✓ Number of attempts made by user                        │
│ ✓ Why escalation was triggered                           │
│ ✓ User's preferred language                              │
│                                                          │
│ Human can then:                                          │
│ • Verify information directly with user                  │
│ • Make final decision on case                            │
│ • Process complaint immediately                          │
│ • Provide personalized resolution                        │
│ • Update case status                                     │
└──────────────────────────────────────────────────────────┘

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TRIGGERS FOR HANDOVER (All lead here):

1. Explicit request: "I want to talk to a human"
2. Attempts exceeded: attempts > 3
3. Escalation critical: score >= 0.75
4. Risk keywords: "assault", "abuse", "threat", etc.
```

DIAGRAM 4: SOFT RESET FLOW

```
                  ╔═══════════════════════════════════╗
                  ║ SOFT RESET TRIGGERED              ║
                  ║ (Line 3575-3583)                  ║
                  ╚═══════════════════════════════════╝
                                 │
                 ┌───────────────┴───────────────┐
                 │                               │
           Trigger 1:                       Trigger 2:
        Score >= 0.55                    Correction Loop
           (User                       (awaiting_correction
```

```
        frustrated)                        + attempts >= 2)
             │                                   │
             └───────────────┬───────────────────┘
                             │
                             ▼
                 ┌───────────────────────────────┐
                 │ state.soft_reset()            │
                 │ (Line 313-325)                │
                 └───────────────┬───────────────┘
                                 │
        ┌────────────────────────┼────────────────────────┐
        │                        │                        │
        ▼                        ▼                        ▼
  ╔══════════════╗        ╔══════════════╗        ╔══════════════╗
  ║ CLEARED      ║        ║ PRESERVED    ║        ║ FLAGS SET    ║
  ║ (Reset)      ║        ║ (Keep Data)  ║        ║              ║
  ╚══════════════╝        ╚══════════════╝        ╚══════════════╝
  │                       │                       │
  ├─ attempts = 0         ├─ entities             ├─ just_reset=True
  ├─ field_attempts={}    ├─ entity_confidence    ├─ awaiting_new=True
  ├─ correction_count=0   ├─ sentiment            └─ awaiting_verify=False
  ├─ correction_passes=0  ├─ language
  └─ fields_corrected={}  ├─ department
                          └─ intent

    EFFECT: User gets fresh start with cleared effort,
           but system remembers extracted data & context

                             │
                             ▼
                 ┌───────────────────────────────────────┐
                 │ ask_rephrase() called                 │
                 │ (Line 3580-3582)                      │
                 │                                       │
                 │ SYSTEM MESSAGE:                       │
                 │ "Let me start fresh. Can you tell me  │
                 │  about your issue again from the      │
                 │  beginning? I have your information   │
                 │  saved (name, email, etc.)"           │
                 └───────────────────┬───────────────────┘
                                     │
                                     ▼
                 ┌───────────────────────────────────────┐
                 │ USER RESPONDS WITH NEW EXPLANATION    │
                 │                                       │
                 │ Example:                              │
                 │ "I ordered a laptop 2 weeks ago and   │
                 │  it arrived with broken screen"       │
                 └───────────────────┬───────────────────┘
                                     │
                                     ▼
                 ┌───────────────────────────────────────┐
                 │ FRESH EXTRACTION                      │
                 │ • extract_full() - new parsing        │
                 │ • Re-detect intent, sentiment         │
                 │ • Generate new confidence scores      │
                 │ • May get different results!          │
                 │                                       │
```

```
│ (Line 3597-3611)                          │
└───────────────────────────────────────────┘
                    │
                    ▼
┌───────────────────────────────────────────┐
│ CHECK: just_reset == True?                │
│ (Line 3617)                               │
│                                           │
│ YES → Reverify low-confidence fields      │
│ (< 0.80 confidence)                       │
└───────────────────────────────────────────┘
                    │
                    ▼
┌───────────────────────────────────────────┐
│ REVERIFICATION                            │
│                                           │
│ "You mentioned 'laptop'. Is that          │
│  the correct product?"                    │
│                                           │
│ User can:                                 │
│ • Confirm: "Yes" → confidence=0.95        │
│ • Correct: "No, it's a phone"             │
│ • Reject: "No" → ask for correct value    │
│                                           │
└───────────────────────────────────────────┘
                    │
                    ▼
┌───────────────────────────────────────────┐
│ CONTINUE NORMAL FLOW                      │
│                                           │
│ • Check missing fields                    │
│ • Ask for any remaining data              │
│ • Build verification summary              │
│                                           │
│ With FRESH PERSPECTIVE from soft reset    │
└───────────────────────────────────────────┘
```

DIAGRAM 5: LOCAL CORRECTION FLOW (During Verification)

```
╔═══════════════════════════════════════════╗
║ VERIFICATION STAGE                        ║
║ System shows summary                      ║
║                                           ║
║ Name: John Smith                          ║
║ Email: john@email.com                     ║
║ Phone: 555-123-4567                       ║
║ Issue: Laptop arrived broken              ║
║                                           ║
║ "Is this correct?"                        ║
╚═══════════════════════════════════════════╝
                    │
  ┌─────────────────┼─────────────────┐
  │                 │                 │
 YES               NO              PARTIAL
  │          (correction        (some fields
  │            needed)          need change)
```

```
                 │                    │                    │
                 ▼                    ▼                    ▼
        ┌──────────────┐     ┌──────────────┐     ┌──────────────┐
        │ COMPLETION   │     │ IDENTIFY     │     │ IDENTIFY     │
        │ Process      │     │ WHICH FIELD  │     │ WHICH FIELDS │
        │ case         │     │ needs        │     │ need change  │
        └──────────────┘     │ correction   │     └──────────────┘
                             └──────────────┘
                                    │
                                    ▼
                     ┌────────────────────────────────────────┐
                     │ DECISION ENGINE: decide(state, text)   │
                     │ (Line 3803-3806)                       │
                     │                                        │
                     │ Analyzes user response:                │
                     │ • "No, my email is john@work.com"      │
                     │ • Identifies field: "email"            │
                     │ • Returns: action="correct",           │
                     │   target_field="email"                 │
                     └────────────────────────────────────────┘
                                    │
                                    ▼
                     ┌────────────────────────────────────────┐
                     │ FIELD CORRECTION HANDLER               │
                     │ (Line 3814-3860)                       │
                     │                                        │
                     │ action == "correct"                    │
                     └────────────────────────────────────────┘
                                    │
                                    ▼
     ┌────────────────────────────────────────────────────────────┐
     │ Step 1: EXTRACT NEW VALUE                                  │
     │ (Line 3825-3828)                                           │
     │                                                            │
     │ extracted = extract(                                       │
     │     text,                                                  │
     │     mode="field",                    ← Extract specific field
     │     target_field="email",            ← Know which field    │
     │     state=state                      ← Full context        │
     │ )                                                          │
     │                                                            │
     │ RESULT: {email: "john@work.com"}                           │
     └────────────────────────────────────────────────────────────┘
                     │
     ┌───────────────┴──────────────────────────┐
     │                                          │
   YES                                          NO
  Value                                     Extraction
  found                                       failed
     │                                          │
     ▼                                          ▼
     ┌─────────────────────────────┐    ┌──────────────────────────┐
     │ Continue                    │    │ Ask again                │
     └─────────────────────────────┘    │                          │
                 │                      │ "What's your email?"     │
                 ▼                      └──────────────────────────┘
     ┌────────────────────────────────────────────────────────────┐
```

```
┌──────────────────────────────────────────────────────────────────┐
│ Step 2: UPDATE CASE STATE                                        │
│ (Line 3833-3837)                                                 │
│                                                                  │
│ state = update_case(                                             │
│     state,                                                       │
│     extracted,                                                   │
│     mode="normal"                 ← Normal correction mode       │
│ )                                                                │
│                                                                  │
│ This will:                                                       │
│ • Update entity: entities["email"] = "john@work.com"             │
│ • Increment correction_count += 1                                │
│ • Set confidence for email field                                 │
│ • Check for correction drift                                     │
└──────────────────────────────────────────────────────────────────┘
                │
                ▼
┌──────────────────────────────────────────────────────────────────┐
│ Step 3: BOOST CONFIDENCE                                         │
│ (Line 3847)                                                      │
│                                                                  │
│ state.increase_confidence(0.02)                                  │
│ • Confidence += 0.02                                             │
│ • Shows system trusts the correction                             │
│ • Caps at 1.0 (100%)                                             │
└──────────────────────────────────────────────────────────────────┘
                │
                ▼
┌──────────────────────────────────────────────────────────────────┐
│ Step 4: RESET ATTEMPT COUNTERS                                   │
│ (Line 3849-3852)                                                 │
│                                                                  │
│ state.reset_attempts()                                           │
│ • Clear attempts counter for this field                          │
│ • Allows clean re-extraction if needed                           │
│ • Prevents artificial "stuck" state                              │
└──────────────────────────────────────────────────────────────────┘
                │
                ▼
┌──────────────────────────────────────────────────────────────────┐
│ Step 5: CHECK FOR MORE CORRECTIONS                               │
│ (Line 3859-3860)                                                 │
│                                                                  │
│ if state.missing_fields:                                         │
│     return ask_missing()                                         │
│ else:                                                            │
│     return build_verification_response()   ← Show summary        │
│                                                                  │
│ OPTIONS:                                                         │
│ • More fields to collect → ask_missing()                         │
│ • All done → build summary (goto verification)                   │
└──────────────────────────────────────────────────────────────────┘
                │
                ▼
┌──────────────────────────────────────────────────────────────────┐
│ Step 6: CHECK CORRECTION DRIFT                                   │
│ (Automatic in apply_control)                                     │
```

```
During this flow, apply_control() checks:
• Same field corrected 3+ times?
  → sentiment = "high"
• Corrections > field count?
  → sentiment = "high"
• Multi-pass drift?
  → sentiment = "high"

If triggered:
→ escalation_score >= 0.55
→ SOFT RESET triggered again
```

```
SHOW UPDATED SUMMARY (if no drift)

Name: John Smith
Email: john@work.com         ← CORRECTED
Phone: 555-123-4567
Issue: Laptop arrived broken

"Is this correct now?"

User can:
• "Yes" → COMPLETION
• "No, X is still wrong" → Another correction
```